Path 2: Cyber Security 101 > Module 3: Windows and AD Fundamentals > Room 3: Windows Fundamentals 3

https://tryhackme.com/room/windowsfundamentals3xzx

Windows Fundamentals 3:
In part 3 of the Windows Fundamentals module, learn about the built-in Microsoft tools that help keep the device secure, such as Windows Updates, Windows Security, BitLocker, and more...

>> Task 1: Introduction

We will continue our journey exploring the Windows operating system.

To summarize the previous two rooms:

- In **Windows Fundamentals 1**, we covered the desktop, the file system, user account control, the control panel, settings, and the task manager.

- In **Windows Fundamentals 2**, we covered various utilities, such as System Configuration, Computer Management, Resource Monitor, etc.

>> Task 2: Windows Updates

Let's start things off with **Windows Update**.

Windows Update is a service provided by Microsoft to provide security updates, feature enhancements, and patches for the Windows operating system and other Microsoft products, such as Microsoft Defender.

*Updates are typically released on the 2nd Tuesday of each month. This day is called **Patch Tuesday**. That doesn't necessarily mean that a critical update/patch has to wait for the next Patch Tuesday to be released. If the update is urgent, then Microsoft will push the update via the Windows Update service to the Windows devices.*

Refer to the following link to see the **Microsoft Security Update Guide** here (https://msrc.microsoft.com/update-guide).

Windows Update is located in Settings.

Tip: Another way to access Windows Update is from the Run dialog box, or CMD, by running the command **control /name Microsoft.WindowsUpdate**.

Q1) There were two definition updates installed in the attached VM. On what date were these updates installed?
Answer: 5/3/2021

>> Task 3: Windows Security

Per Microsoft, "**Windows Security** is your home to manage the tools that protect your device and your data".

In case you missed it, **Windows Security** is also available in **Settings**.

Focus your attention on **Protection areas**.
:
- **Virus & threat protection**
- **Firewall & network protection**
- **App & browser control**
- **Device security**

Before proceeding, let's provide a quick comment on the status icons.
:
- **Green** means your device is sufficiently protected, and there aren't any recommended actions.
- **Yellow** means there is a safety recommendation for you to review.
- **Red** is a warning that something needs your immediate attention.

Q1) Checking the Security section on your VM, which area needs immediate attention?
Answer: Virus & Threat Protection

>> Task 4: Virus & Threat Protection

Virus & threat protection is divided into two parts:
- **Current threats**
- **Virus & threat protection settings**

- Current threats

Scan options
:
- **Quick scan** - Checks folders in your system where threats are commonly found.
- **Full scan** - Checks all files and running programs on your hard disk. This scan could take longer than one hour.
- **Custom scan** - Choose which files and locations you want to check.

Threat history
:
- **Last scan** - Windows Defender Antivirus automatically scans your device for viruses and other threats to help keep it safe.
- **Quarantined threats** - Quarantined threats have been isolated and prevented from running on your device. They will be periodically removed.
- **Allowed threats** - Allowed threats are items identified as threats, which you allowed to run on your device.

Warning: Allow an item to run that has been identified as a threat only if you are **100%** sure of what you are doing.

Next is *Virus & threat protection settings*.

Virus & threat protection settings

Manage settings
:
- **Real-time protection** - Locates and stops malware from installing or running on your device.

- **Cloud-delivered protection** - Provides increased and faster protection with access to the latest protection data in the cloud.

- **Automatic sample submission** - Send sample files to Microsoft to help protect you and others from potential threats.

- **Controlled folder access** - This feature, if enabled, protects files, folders, and memory areas on your device from unauthorized changes by malicious or unknown applications. If it is enabled, only approved and trusted apps would be allowed to modify the files in the protected folders. To enable this feature, click on the Manage **controlled folder access** button under **Controlled Folder Access** and turn it on.

- **Exclusions** - Windows Defender Antivirus allows you to exclude any files or folders from the antivirus scanning. This is done to reduce the number of false positives. Administrators might not want the antivirus to scan specific files or folders. By adding them to the exclusions list, the antivirus would ignore them and scan all the other files and folders. To add any file or folder to the Windows Defender exclusion list, click on the **Add or remove exclusions** button under **Exclusions** and add as many exclusions as you want.

- **Notifications** - Windows Defender Antivirus will send notifications with critical information about the health and security of your device.

Warning: Excluded items could contain threats that make your device vulnerable. Only use this option if you are 100% sure of what you are doing.

Virus & threat protection updates
:
Check for updates - Manually check for updates to update Windows Defender Antivirus definitions.

Ransomware protection
:
Controlled folder access - Ransomware protection requires this feature to be enabled, which in turn requires Real-time protection to be enabled.

Tip: You can perform on-demand scans on any file/folder by right-clicking the item and selecting 'Scan with Microsoft Defender'.

Q1) Specifically, what is turned off that Windows is notifying you to turn on?
Answer: Real-time protection

>> Task 5: Firewall & Network Protection

What is a **firewall**?

Per Microsoft, "Traffic flows into and out of devices via what we call ports. A firewall is what controls what is - and more importantly isn't - allowed to pass through those ports. You can think of it like a security guard standing at the door, checking the ID of everything that tries to enter or exit".

What is the difference between the 3 (**Domain**, **Private**, and **Public**)?

Per Microsoft, "Windows Firewall offers three firewall profiles: domain, private and public".

- **Domain** - The domain profile applies to networks where the host system can authenticate to a domain controller.
- **Private** - The private profile is a user-assigned profile and is used to designate private or home networks.
- **Public** - The default profile is the public profile, used to designate public networks such as Wi-Fi hotspots at coffee shops, airports, and other locations.

Warning: Unless you are **100%** confident in what you are doing, it is recommended that you leave your Windows Defender Firewall enabled.

🔹 Allow an app through firewall (**Control Panel\System and Security\Windows Defender Firewall\Allowed apps**):

You can view what the current settings for any firewall profile are. Several apps have access in the Private and/or Public firewall profile. Some of the apps will provide additional information if it's available via the **Details** button.

🔹 Advanced Settings (**Control Panel\System and Security\Windows Defender Firewall**) / (**Win+R**: WF.msc):

Configuring the **Windows Defender Firewall** is for advanced Windows users. Refer to the following Microsoft documentation on best practices here (https://learn.microsoft.com/en-us/windows/security/operating-system-security/network-security/windows-firewall/configure).

Tip: Command to open the Windows Defender Firewall is WF.msc.

Q1) If you were connected to airport Wi-Fi, what most likely will be the active firewall profile?
Answer: Public Network

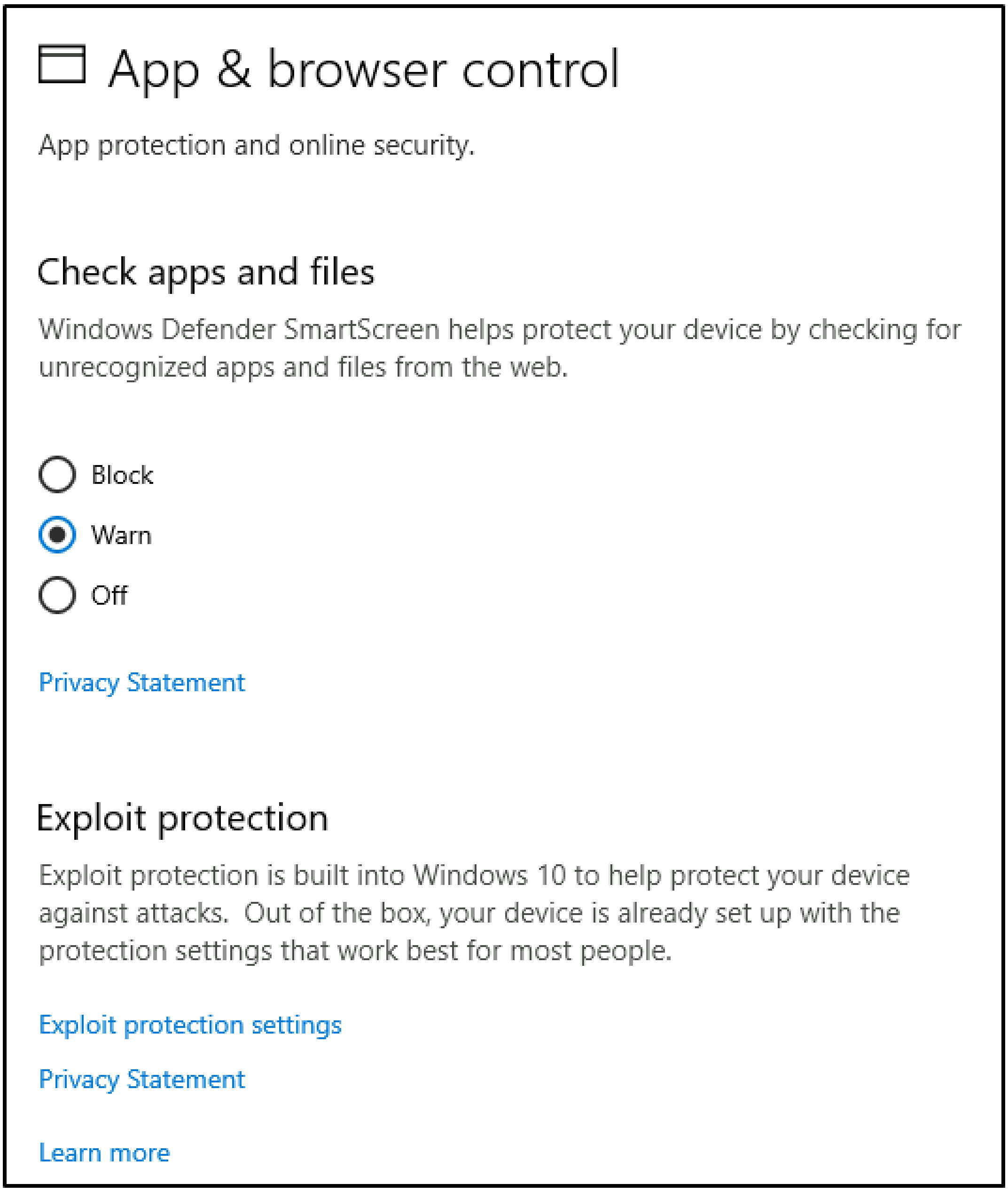
>> Task 6: App & Browser Control

In this section, you can change the settings for the **Microsoft Defender SmartScreen.**

Per Microsoft, "Microsoft Defender SmartScreen protects against phishing or malware websites and applications, and the downloading of potentially malicious files".

Refer to the official Microsoft document for more information on Microsoft Defender SmartScreen here
(https://feedback.smartscreen.microsoft.com/smartscreenfaq.aspx).

You can see the status of the smart screen set to **Warn** below. You can also change it to either **Block** or completely turned **Off**.



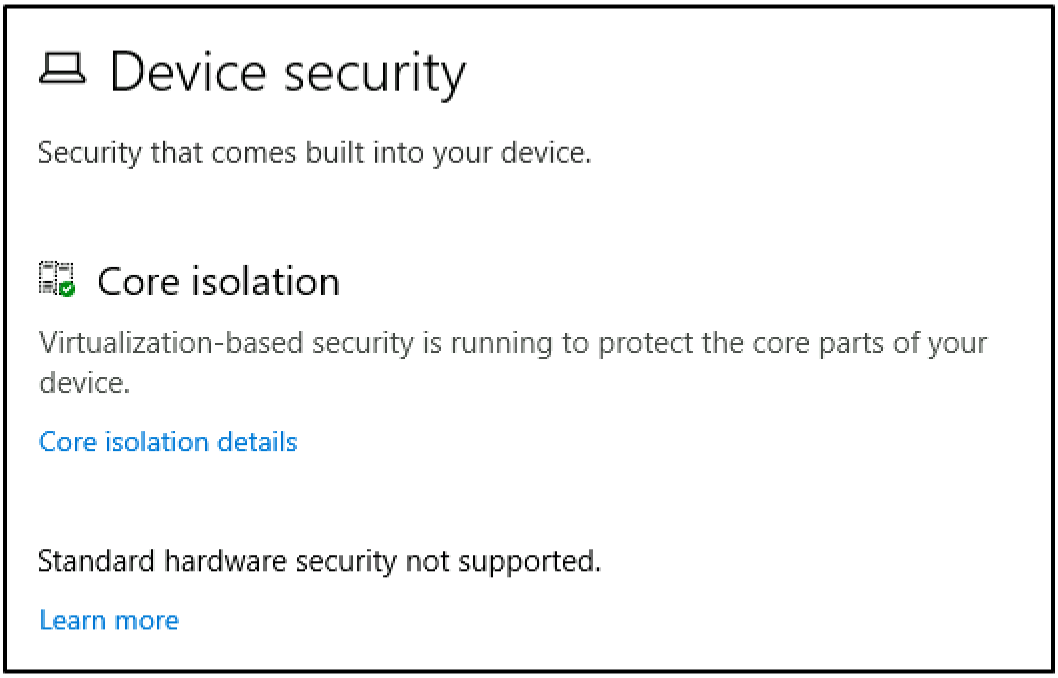
Check apps and files
:
Windows Defender SmartScreen helps protect your device by checking for unrecognized apps and files from the web.

Exploit protection
:
Exploit protection is built into Windows 10 (and, in our case, Windows Server 2019) to help protect your device against attacks.

Warning: Unless you are **100%** confident in what you are doing, it is recommended that you leave the default settings.

>> Task 7: Device Security

Even though you'll probably never change any of these settings, for completion's sake, it will be covered briefly.

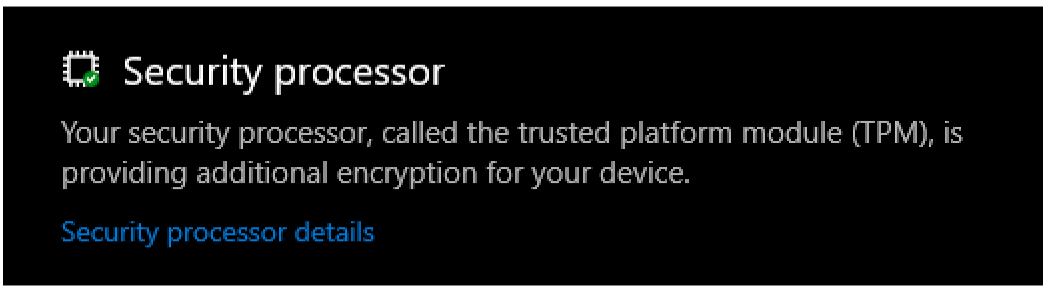


Core isolation
:
Memory Integrity - Prevents attacks from inserting malicious code into high-security processes.

Warning: Unless you are **100%** confident in what you are doing, it is recommended that you leave the default settings.

The below images are from another machine to show another security feature that should be available in a personal Windows 10 device.

Security processor
:



What is the **Trusted Platform Module (TPM)**?
:
Per Microsoft, "Trusted Platform Module (TPM) technology is designed to provide hardware-based, security-related functions. A TPM chip is a secure crypto-processor that is designed to carry out cryptographic operations. The chip includes multiple physical security mechanisms to make it tamper-resistant, and malicious software is unable to tamper with the security functions of the TPM".

Q1) What is the TPM?
Answer: Trusted Platform Module

>> Task 8: BitLocker

What is **BitLocker**?

Per Microsoft, "BitLocker Drive Encryption is a data protection feature that integrates with the operating system and addresses the threats of data theft or exposure from lost, stolen, or inappropriately decommissioned computers".

On devices with TPM installed, BitLocker offers the best protection.

Per Microsoft, "BitLocker provides the most protection when used with a Trusted Platform Module (TPM) version 1.2 or later. The TPM is a hardware component installed in many newer computers by the computer manufacturers. It works with BitLocker to help protect user data and to ensure that a computer has not been tampered with while the system was offline".

Refer to the official Microsoft documentation to learn more about BitLocker here (https://learn.microsoft.com/en-us/windows/security/operating-system-security/data-protection/bitlocker/).

Q1) We should use a removable drive on systems **without** a TPM version 1.2 or later. What does this removable drive contain?
Answer: Startup Key

>> Task 9: Volume Shadow Copy Service

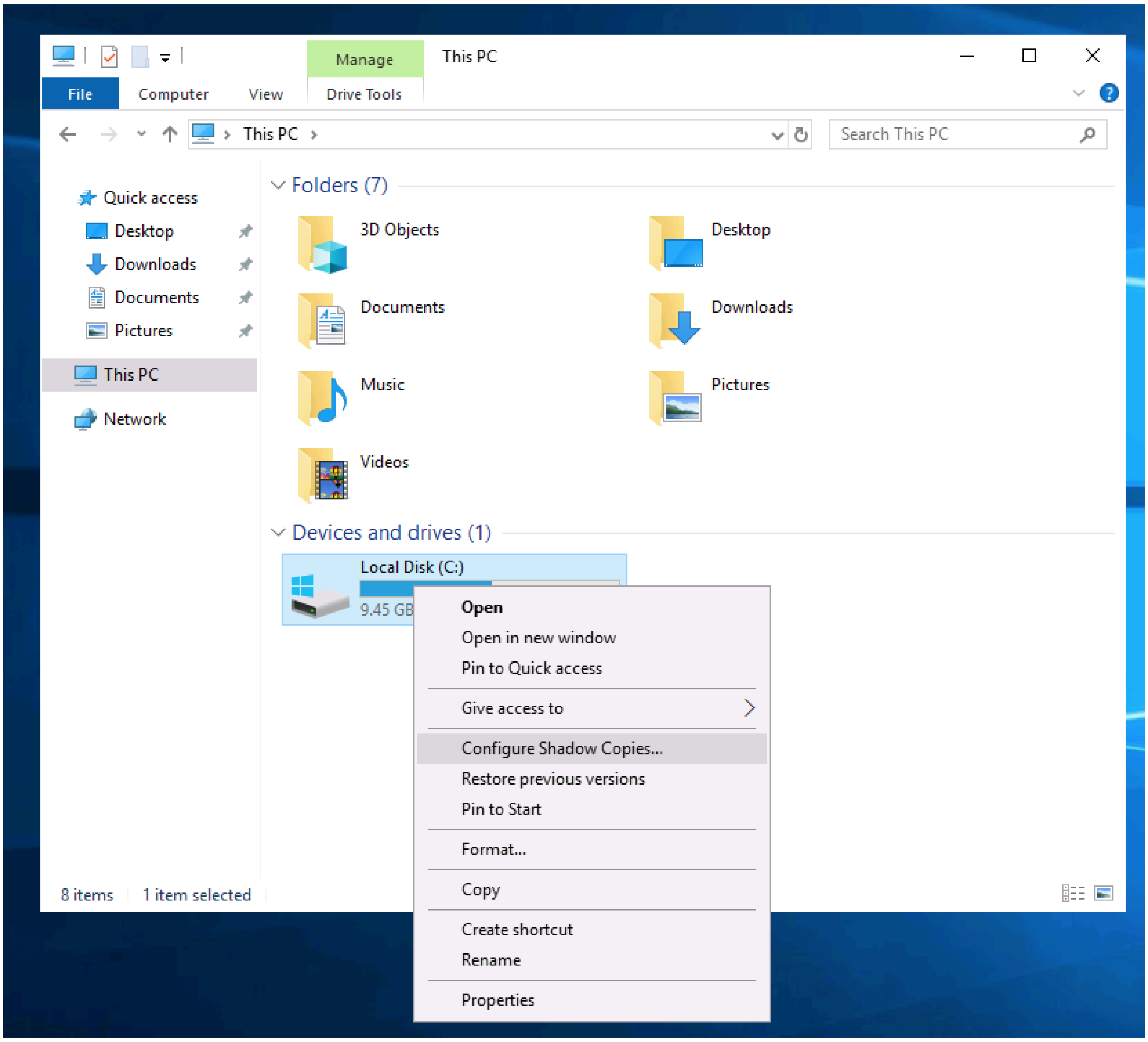
Per Microsoft, the **Volume Shadow Copy Service** (VSS) coordinates the required actions to create a consistent shadow copy (also known as a snapshot or a point-in-time copy) of the data that is to be backed up.

Volume Shadow Copies are stored on the System Volume Information folder on each drive that has protection enabled.

If VSS is enabled (**System Protection** turned on), you can perform the following tasks from within **advanced system settings**.

- Create a restore point
- Perform system restore
- Configure restore settings
- Delete restore points

From a security perspective, malware writers know of this Windows feature and write code in their malware to look for these files and delete them. Doing so makes it impossible to recover from a ransomware attack unless you have an offline/off-site backup.



Bonus: If you wish to interact hands-on with VSS, I suggest exploring Day 23 of Advent of Cyber 2 (https://tryhackme.com/room/adventofcyber2).

Q1) What is VSS?
Answer: Volume Shadow Copy Service

>> Task 10: Conclusion

To learn more about the Windows OS, you'll need to continue the journey on your own.

Further reading material:

- Antimalware Scan Interface (https://learn.microsoft.com/en-us/windows/win32/amsi/antimalware-scan-interface-portal)

- Credential Guard (https://learn.microsoft.com/en-us/windows/security/identity-protection/credential-guard/configure?tabs=intune)

- Windows 10 Hello (https://support.microsoft.com/en-us/windows/configure-windows-hello-dae28983-8242-bb2a-d3d1-87c9d265a5f0#:~:text=Windows%2010,in%20with%20just%20your%20PIN.)

- CSO Online - The best new Windows 10 security features (https://www.csoonline.com/article/564531/the-best-new-windows-10-security-features.html)

Note: Attackers use built-in Windows tools and utilities in an attempt to go undetected within the victim environment. This tactic is known as Living Off The Land. Refer to the following resource here (https://lolbas-project.github.io/) to learn more about this.